

Framing Against Itself: Intra-Textual Tensions in NYT Iran Coverage

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1. Secondary Literature

Computational framing studies overwhelmingly treat news articles as monolithic carriers of a single frame. Yet a newspaper is not a single voice: its headlines, leads, body text, editorials, and cultural pages are produced under different constraints, for different audiences, with different rhetorical goals. Our overarching question asks: *does the editorial apparatus of the New York Times construct a coherent image of Iran—or do its structural components pull in systematically contradictory directions, and if so, how has this tension evolved over five decades of crisis coverage?*

This question builds on three bodies of literature. First, framing theory provides the conceptual foundation. Entman (1993) defined framing as selecting “some aspects of a perceived reality” and making them salient, identifying four frame functions: problem definition, causal attribution, moral evaluation, and treatment recommendation. Pan and Kosicki (1993) decomposed news discourse into four structural dimensions—syntactical (headline, lead), script (narrative), thematic (causal reasoning), and rhetorical (stylistic)—calling headlines “the most salient cue” for reader interpretation. Crucially, these dimensions can carry *different* frame signals, but subsequent computational work has rarely measured the divergence between them. Experimental evidence suggests such divergence matters: Ecker et al. (2014) showed that misleading headlines bias inferential reasoning even after participants read the full text—a “continued influence effect” that persists because the headline establishes the initial interpretive frame. Lee and Maslog (2005) found that shorter news formats systematically favour war journalism over peace journalism, suggesting a structural mechanism by which compressed elements—headlines, briefs—amplify conflict framing relative to longer-form bodies.

Second, scholarship on Western media framing of Iran provides the substantive case. Said’s *Covering Islam* (1981) argued qualitatively that American media reduce Iran to a shifting set of security anxieties. Izadi and Saghaye-Biria (2007) extended this through discourse analysis of 29 editorials across the *NYT*, *Washington Post*, and *Wall Street Journal* (1984–2004), identifying six recurring Orientalist themes—“Oriental untrustworthiness,” “Islam as a threat,” “Oriental irrationality”—structuring coverage of Iran’s nuclear program. Crucially, their analysis was confined to editorials; how these frames interact with news reporting in the same newspaper remains unexplored. More recently, Guo et al. (2025) applied the war/peace journalism framework to nuclear weapons coverage in the *NYT*, *Guardian*, and *Global Times*, finding the *NYT* predominantly war-journalism oriented—a finding our corpus allows us to test longitudinally across five decades rather than a single conflict.

Third, the genre and section structure of a newspaper creates a natural experiment. Entman (2004) showed that frames cascade from government elites through media to the public, explicitly distinguishing editorial pages from news pages as carrying different frame-diffusion capacities. If the same newspaper simultaneously produces a threatening Iran in its World section (threat/humanizing ratio of 0.33 in our preliminary data) and a culturally rich Iran in its Arts section (ratio: 4.54), the institution is not delivering a single frame but an internally contradictory discourse—one that can be mapped computationally. Where image metadata is available (21% of articles, concentrated post-2004), we extend this analysis to visual-textual tensions. Preliminary results show that imaged articles carry higher threat density

than non-imaged ones (14.6 vs. 12.1 per 1,000 words), and that 61% of diplomacy-keyword articles simultaneously carry security keywords—suggesting that even diplomatic coverage is visually and semantically co-framed with threat.

2. Methods

2.1. Data Enrichment

Our first report noted that the NYT Archive API returns only metadata, abstracts, and lead paragraphs—a significant limitation for intra-textual analysis. We have since addressed this by building a scraping pipeline that retrieves full article bodies from the Internet Archive’s Wayback Machine and extracts text via `trafilatura` (Barbarese 2021). This enriched the corpus with 29,837 full articles (68.9% of 43,316), yielding a $4.4\times$ increase in total text—from 4.5 million to 20.6 million words. The enriched dataset is publicly available.¹

Limitations. Coverage is temporally uneven: strongest in the 1990s (94.4%) and weakest in the 2020s (3.1%), reflecting Wayback Machine archival patterns rather than editorial selection. Pre-1981 articles (26.3% coverage) are often behind the NYT’s scanned-page paywall. We address this bias by reporting decade-stratified results and by retaining abstracts and lead paragraphs as a fallback for articles without full text—the headline–abstract comparison remains available for the full 43,316-article corpus regardless of full-text availability.

2.2. Lexicon-Based Framing Analysis

We operationalize framing tension through three curated lexicons—*threat* (63 terms: “attack,” “nuclear,” “sanctions,” “kill,” etc.), *diplomacy* (34 terms: “negotiate,” “agreement,” “reform,” etc.), and *humanizing* (44 terms: “family,” “artist,” “freedom,” “culture,” etc.)—mapped to the war/peace journalism indicators of Lee and Maslog (2005). Replicating their framework on our corpus, we find that headlines carry a war/peace ratio of 2.71 versus 2.18 in full text, confirming their finding that compressed formats amplify war framing; the ratio has declined from 2.68 (1980s) to 1.72 (2020s), suggesting a slow structural shift. For each article, we compute lexicon density (hits per 1,000 tokens) separately for each structural component: headline, abstract, lead paragraph, and full text. The *framing tension score* is defined as the divergence between component-level densities—for example, the headline–body threat gap ($D_{HL} - D_{FT}$).

We choose lexicon-based methods over transformer classifiers at this stage for transparency (every decision is traceable to a word match, enabling qualitative validation) and scalability (29,837 articles process in under two minutes). We plan to complement this with BERTopic dynamic topic modeling (Grootendorst 2022) and transformer-based frame classifiers following Kwak, An, and Ahn (2020), who fine-tuned BERT on Card et al.’s (2015) 15-frame typology to classify 1.5 million NYT articles. The lexicon approach establishes baselines that the transformer models will refine.

2.3. Preliminary Validation

Initial results confirm that intra-textual framing tensions are measurable and non-trivial. Headlines carry $2.1\times$ the threat density of their own article bodies (28.5 vs. 13.4 per 1,000 words), and this amplification is growing: from 12% of articles in the 1970s to 42% in the 2020s. Humanizing language is the only lexicon category where full text exceeds headlines—suggesting that the editorial apparatus systematically concentrates threat in high-salience positions while distributing humanization across the less-read body. Across genres, reviews are $7\times$ more humanizing than news analysis articles within the same newspaper. These patterns are robust across the decades with strong coverage (1980s–2010s) and—to our knowledge—have not been measured at this scale for Middle Eastern coverage.

¹<https://huggingface.co/datasets/alunxu/nyt-iran-coverage>

References

- Barbareasi, Adrien. 2021. "Trafilatura: A Web Scraping Library and Command-Line Tool for Text Discovery and Extraction." *Proceedings of the ACL System Demonstrations*, 122–131.
- Card, Dallas, Amber E. Boydston, Justin H. Gross, Philip Resnik, and Noah A. Smith. 2015. "The Media Frames Corpus: Annotations of Frames Across Issues." In *Proceedings of ACL-IJCNLP 2015*, 438–444.
- Ecker, Ullrich K. H., Stephan Lewandowsky, Ee Pin Chang, and Rekha Pillai. 2014. "The Effects of Subtle Misinformation in News Headlines." *Journal of Experimental Psychology: Applied* 20 (4): 323–335. <https://doi.org/10.1037/xap0000028>.
- Entman, Robert M. 1993. "Framing: Toward Clarification of a Fractured Paradigm." *Journal of Communication* 43 (4): 51–58. <https://doi.org/10.1111/j.1460-2466.1993.tb01304.x>.
- Entman, Robert M. 2004. *Projections of Power: Framing News, Public Opinion, and U.S. Foreign Policy*. Chicago: University of Chicago Press.
- Grootendorst, Maarten. 2022. "BERTopic: Neural Topic Modeling with a Class-Based TF-IDF Procedure." In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*.
- Guo, Yijia, Shuang Liu, Yuqi Shao, and Crispin Maslog. 2025. "Valuing or Devaluing Nuclear Weapons in the War Journalism: A Cross-National Comparative Content Analysis." *International Communication Gazette*, <https://doi.org/10.1177/17480485241311544>.
- Izadi, Foad, and Hakimeh Saghaie-Biria. 2007. "A Discourse Analysis of Elite American Newspaper Editorials: The Case of Iran's Nuclear Program." *Journal of Communication Inquiry* 31 (2): 140–165. <https://doi.org/10.1177/0196859906298073>.
- Kwak, Haewoon, Jisun An, and Yong-Yeol Ahn. 2020. "A Systematic Media Frame Analysis of 1.5 Million New York Times Articles from 2000 to 2017." In *Proceedings of the 12th ACM Conference on Web Science*, 305–314. <https://doi.org/10.1145/3394231.3397921>.
- Lee, Seow Ting, and Crispin C. Maslog. 2005. "War or Peace Journalism? Asian Newspaper Coverage of Conflicts." *Journal of Communication* 55 (2): 311–329. <https://doi.org/10.1111/j.1460-2466.2005.tb02674.x>.
- Pan, Zhongdang, and Gerald M. Kosicki. 1993. "Framing Analysis: An Approach to News Discourse." *Political Communication* 10 (1): 55–75. <https://doi.org/10.1080/10584609.1993.9962963>.
- Said, Edward W. 1981. *Covering Islam: How the Media and the Experts Determine How We See the Rest of the World*. New York: Pantheon Books.